Problem set 9 - Due: Monday, March 23

Solve the following ODEs using variation of parameters, given the complementary function y_c

1.
$$x^2y'' - 4xy' + 6y = x^3, \quad y_c = c_1x^2 + c_2x^3.$$

We have

$$y'' - \frac{4}{x}y' + \frac{6}{x^2}y = x.$$

So, with

$$W = x^2(3x^2) - x^3(2x) = 3x^4 - 2x^4 = x^4,$$

we have the particular solution

$$\begin{aligned} y_p &= -x^2 \int \frac{x^3 x}{x^4}\ dx + x^3 \int \frac{x^2 x}{x^4}\ dx \\ &= -x^2 x + x^3 \ln|x| = x^3 \ln|x| - x^3. \end{aligned}$$

Thus, the solution is

$$y = c_1x^2 + c_2x^3 + x^3\ln|x| - x^3 = c_1x^2 + c_2x^3 + x^3\ln|x|.$$

2.
$$4x^2y'' - 4xy' + 3y = 8x^{\frac{4}{3}}, \quad y_c = c_1x^{\frac{1}{2}} + c_2x^{\frac{3}{2}}.$$

We have

$$y'' - \frac{1}{x}y' + \frac{3}{4x^2}y = 8x^{\frac{4}{3}-2} = 8x^{-\frac{2}{3}}.$$

With

$$W = x^{\frac{1}{2}}\left(\frac{3}{2}x^{\frac{1}{2}}\right) - x^{\frac{3}{2}}\left(\frac{1}{2}x^{-\frac{1}{2}}\right) = \frac{3}{2}x - \frac{1}{2}x = x,$$

we get the particular solution

$$\begin{aligned} y_p &= -x^{\frac{1}{2}} \int \frac{x^{\frac{3}{2}}8x^{-\frac{2}{3}}}{x}\ dx + x^{\frac{3}{2}} \int \frac{x^{\frac{1}{2}}8x^{-\frac{2}{3}}}{x}\ dx \\ &= -x^{\frac{1}{2}} \int 8x^{-\frac{1}{6}}\ dx + x^{\frac{3}{2}} \int 8x^{-\frac{7}{6}}\ dx \\ &= -x^{\frac{1}{2}}\left(\frac{48}{5}\right)x^{\frac{5}{6}} + x^{\frac{3}{2}}(-48)x^{-\frac{1}{6}} \\ &= -\frac{48}{5}x^{\frac{8}{6}} - 48x^{\frac{8}{6}} = -\frac{6}{5}(48)x^{\frac{8}{6}} = -\frac{288}{5}x^{\frac{4}{3}}. \end{aligned}$$

Thus, the solution is

$$y = c_1x^{\frac{1}{2}} + c_2x^{\frac{3}{2}} - \frac{288}{5}x^{\frac{4}{3}}.$$

3.

$$x^2y'' - 2xy' + 2y = x^{\frac{9}{2}}, \quad y_c = c_1x + c_2x^2.$$

We have

$$y'' - \frac{2}{x}y' + \frac{2}{x^2}y = x^{\frac{5}{2}}.$$

Now, since

$$W = x(2x) - x^2 = 2x^2 - x^2 = x^2,$$

we have the particular solution

$$\begin{aligned}
y_p &= -x\int \frac{x^2x^{\frac{5}{2}}}{x^2}\,dx + x^2\int \frac{xx^{\frac{5}{2}}}{x^2}\,dx \\
&= -x\int x^{\frac{5}{2}}\,dx + x^2\int x^{\frac{3}{2}}\,dx \\
&= -\frac{2}{7}xx^{\frac{7}{2}} + \frac{2}{5}x^2x^{\frac{5}{2}} = -\frac{2}{7}x^{\frac{9}{2}} + \frac{2}{5}x^{\frac{9}{2}} = \frac{4}{35}x^{\frac{9}{2}}.
\end{aligned}$$

Thus, the solution is

$$y = c_1x + c_2x^2 + \frac{4}{35}x^{\frac{9}{2}}.$$

4.

$$x^2y'' - xy' - 3y = x^{\frac{3}{2}}, \quad y_c = c_1\frac{1}{x} + c_2x^3.$$

We have

$$y'' - \frac{1}{x}y' - \frac{3}{x^2}y = x^{-\frac{1}{2}}.$$

So, with

$$W = \frac{1}{x}(3x^2) - \left(-\frac{1}{x^2}x^3\right) = 3x + x = 4x,$$

we have

$$\begin{aligned}
y_p &= -\frac{1}{x}\int \frac{x^3x^{-\frac{1}{2}}}{4x}\,dx + x^3\int \frac{\frac{1}{x}x^{-\frac{1}{2}}}{4x}\,dx \\
&= -\frac{1}{x}\int \frac{1}{4}x^{\frac{3}{2}}\,dx + x^3\int \frac{1}{4}x^{-\frac{5}{2}}\,dx \\
&= -\frac{1}{4x}\left(\frac{2}{5}\right)x^{\frac{5}{2}} + \frac{x^3}{4}\left(-\frac{2}{3}\right)x^{-\frac{3}{2}} \\
&= -\frac{1}{10}x^{\frac{3}{2}} - \frac{1}{6}x^{\frac{3}{2}} = -\frac{4}{15}x^{\frac{3}{2}}.
\end{aligned}$$

So,

$$y = c_1\frac{1}{x} + c_2x^3 - \frac{4}{15}x^{\frac{3}{2}}.$$